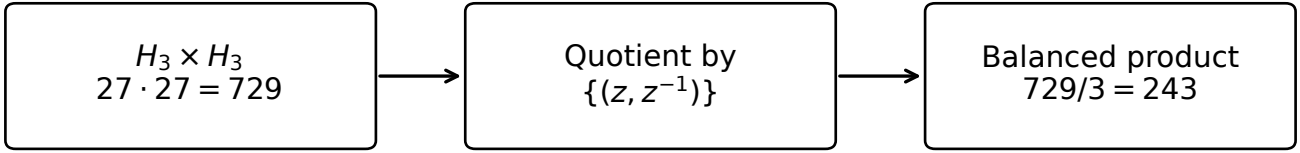

One shared center
not two unrelated centers

Heisenberg degree
$3 \cdot 3 = 9$

Orthogonal factors commute; the actual extensions are inputs.